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**Project Based Learning Report
On**

**“Region Splitting and Merging for Terrain Classification
Quadtree segmentation”**

Subject: - Image Processing

Semester/Year: III/V

Details:

Sr. No	Name of Students & USN	Problem Understanding & Objective[2M]	Implementation & Functionality [4M]	Output Quality & Analysis [2M]	Report/ Presentation Quality [2M]	Total [10M]
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Date of Submission:

1. ABSTRACT

Terrain classification is useful for understanding different areas in a satellite or remote sensing image, such as water, vegetation, buildings, and open land. In this project, we use **region splitting and merging with quadtree segmentation** to divide the image into smaller regions and identify areas having similar characteristics. The image is first split into smaller parts when a region is not uniform, and similar neighboring regions are then merged together. This makes the image easier to analyze and helps in separating different types of terrain. The project shows how a simple region-based image processing technique can be applied to remote sensing images for terrain classification.

2. INTRODUCTION

Remote sensing images provide useful information about the Earth's surface and are widely used to study different types of terrain. A single satellite image may contain several features such as vegetation, water bodies, roads, buildings, and bare land. Identifying these areas manually can be difficult, especially when the image is large or contains many different regions.

Image segmentation is used to divide an image into smaller and meaningful regions. In this project, we use **Quadtree Segmentation with Region Splitting and Merging**. The image is first divided into four smaller regions whenever a region is not sufficiently uniform. This process continues until the regions satisfy the selected condition. After splitting, regions with similar characteristics are merged to form larger meaningful areas.

The main purpose of this project is to understand how region-based segmentation can be applied to remote sensing images for terrain classification. The approach provides a simple way to separate different terrain regions and makes the resulting image easier to analyze.

3. AIM AND OBJECTIVES

Aim:

To develop a region-based image segmentation system using Quadtree Segmentation with Region Splitting and Merging for classifying different terrain regions in remote sensing images.

Objectives:

1. To study the concept of region splitting and merging in image segmentation.
2. To implement Quadtree decomposition for dividing a terrain image into smaller regions.
3. To identify homogeneous regions based on pixel characteristics.
4. To merge neighboring regions having similar properties.
5. To separate and visualize different terrain classes such as water, vegetation, urban areas, and bare land.

4. PROBLEM STATEMENT

Remote sensing images contain different types of terrain, and these areas may have different colors, textures, and intensity values. Identifying these regions directly from a large satellite image can be difficult because the image may contain complex and mixed areas.

The problem is to develop an image segmentation method that can divide a remote sensing image into meaningful regions and group similar areas together. In this project, **Quadtree Segmentation** is used to recursively split non-uniform regions into smaller parts. After splitting, similar neighboring regions are merged to reduce unnecessary divisions and obtain more meaningful terrain regions.

The proposed approach aims to provide a simple and understandable method for segmenting remote sensing images into terrain classes such as **vegetation, water, urban areas, and bare land**.

5. LITERATURE REVIEW

Previous research shows that region-based segmentation is useful for analyzing remote sensing images. Quadtree methods help divide an image into smaller regions, while region-merging techniques combine similar regions to produce more meaningful areas. The reviewed studies also show that the choice of similarity and homogeneity criteria affects the quality of segmentation. Based on these ideas, this project combines quadtree region splitting and merging to classify different terrain regions in a simple and understandable way.

Sr. No.	Author(s) & Year	Method / Approach	Key Findings	Relevance to Proposed Project
1	Fisher & Pathirana (1991)	Quadtree-based spatial and spectral classification	Used quadtree segmentation with spectral and topographic information for remote sensing image classification.	Directly supports the use of quadtree segmentation for terrain classification.
2	Wuest & Zhang (2009)	Split-and-Merge Image Segmentation	Studied a region-based method in which an image is divided into smaller regions and similar regions are merged.	Provides the basic concept of region splitting and merging used in this project.
3	Yang et al. (2014)	Hybrid Region Merging	Used region-merging techniques to improve the segmentation of high-resolution remote sensing images.	Supports the use of region merging to obtain meaningful terrain regions.
4	Yang et al. (2021)	Multiple-Criteria Region Merging	Studied different criteria for merging image regions and their effect on segmentation accuracy.	Helps in selecting suitable similarity and homogeneity conditions for merging regions.

6. PROBLEM DEFINATION & METHODOLOGY

Problem Definition

Remote sensing images contain different types of terrain, but the boundaries between these regions are not always clear. Areas such as vegetation, water, urban land, and bare soil can have similar pixel values in some parts of an image. Because of this, simply dividing the image based on individual pixels may not give accurate or meaningful regions.

The problem addressed in this project is to segment a remote sensing image into meaningful terrain regions using region splitting and merging. A quadtree structure is used to divide the image into smaller regions based on their homogeneity. Regions that have similar properties are then merged to form larger and more meaningful areas. The final segmented image is used to visualize the different terrain classes.

Methodology

The proposed methodology consists of the following steps:

1. **Input Image:**

A remote sensing or satellite image is taken as the input for the project.

2. **Preprocessing:**

The input image is resized if required and converted into a suitable format for processing. Basic preprocessing helps maintain consistent image dimensions and makes further processing easier.

3. **Region Splitting:**

The complete image is initially considered as one region. Its pixel values are checked to determine whether the region is sufficiently homogeneous. If the region is not homogeneous, it is divided into four smaller regions using the quadtree approach. This process is repeated for regions that still do not satisfy the homogeneity condition.

4. **Homogeneity Check:**

For each region, a simple statistical measure such as the variation or standard deviation of pixel values is calculated. A threshold is used to decide whether the region should be split further.

5. **Region Merging:**

After the splitting process, neighboring regions are compared. Regions having similar pixel characteristics are merged together. This reduces unnecessary small regions and produces larger meaningful areas.

6. **Terrain Classification:**

The resulting regions are assigned to terrain categories such as vegetation, water, urban areas, and bare land based on their image characteristics.

7. **Output:**

The final result is displayed as a segmented terrain map. The original image, split regions, and final merged/classified image can be compared to understand the performance of the method.

7. MATHEMATICAL EQUATION/MODELING

The proposed method uses basic statistical measures to decide whether a region should be split or merged. The main measures used are the **mean**, **variance**, and **standard deviation** of the pixel values in each region.

7.1 Mean of a Region

For a region containing N pixels, the mean pixel value is calculated as:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

where:

- μ = mean pixel value of the region
- N = total number of pixels
- x_i = intensity value of the i^{th} pixel

The mean represents the average intensity of the region.

7.2 Variance

Variance is used to measure how much the pixel values differ from the mean:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

where σ^2 represents the variance of the region.

A low variance indicates that the pixels are relatively similar, while a high variance indicates that the region contains more variation.

7.3 Standard Deviation

The standard deviation is the square root of variance:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

A region with a small standard deviation is considered more homogeneous. A region with a large standard deviation is considered non-homogeneous.

7.4 Region Splitting Condition

Let T_s be the splitting threshold. A region is split into four smaller regions when:

$$\sigma > T_s$$

If:

$$\sigma \leq T_s$$

the region is considered sufficiently homogeneous and no further splitting is performed.

The quadtree structure therefore follows:

$$R \rightarrow \{R_1, R_2, R_3, R_4\}$$

where R is the original region and R_1, R_2, R_3, R_4 are its four smaller sub-regions.

7.5 Region Merging Condition

After splitting, neighboring regions are compared using their mean pixel values. Two neighboring regions R_i and R_j can be merged when their mean values are sufficiently close:

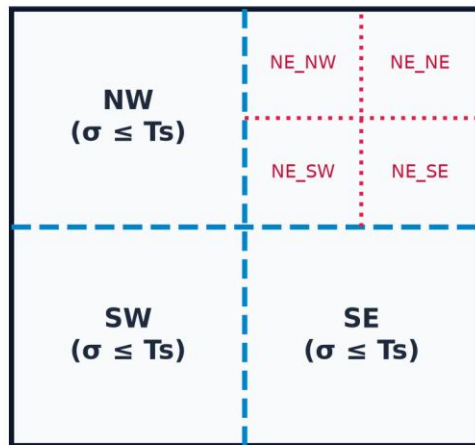
$$|\mu_i - \mu_j| \leq T_m$$

where:

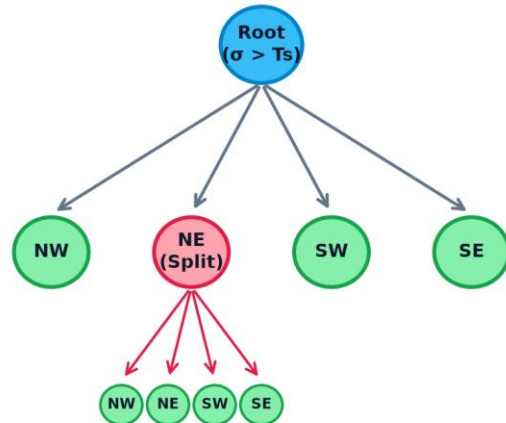
- μ_i = mean value of region R_i
- μ_j = mean value of region R_j
- T_m = merging threshold

If the difference between the two mean values is within the selected threshold, the regions are considered similar and are merged.

$$R_{ij} = R_i \cup R_j$$



(a) Spatial Quadrant Decomposition



(b) Hierarchical Quaternary Tree

Figure 7.1. Quadtree spatial decomposition and quaternary tree representation.

7.6 Overall Decision Model

The complete decision process can be represented as:

If $\sigma > T_s \Rightarrow$ Split the region

If $\sigma \leq T_s \Rightarrow$ Keep the region

and for neighboring regions:

If $|\mu_i - \mu_j| \leq T_m \Rightarrow$ Merge the regions

Thus, the mathematical model provides a simple basis for deciding which regions need further division and which similar regions can be combined. The threshold values T_s and T_m can be adjusted according to the characteristics of the input remote sensing image.

8. PSEUDO CODE/ALGORITHM

START

1. Read the remote sensing image.
2. Preprocess and resize the image if required.
3. Select splitting threshold (T_s) and merging threshold (T_m).
4. Calculate the mean and standard deviation of the image region.
5. IF standard deviation $> T_s$:
 - Divide the region into four smaller regions.
 - Repeat the splitting process for each region.
- ELSE:
 - Keep the region unchanged.
6. Compare neighboring regions.
7. IF difference between their mean values $\leq T_m$:
 - Merge the regions.
8. Assign terrain classes to the final regions.
9. Display the segmented terrain image.

END

9. IMPLEMENTATION

The proposed method was implemented using **Python**, with OpenCV and NumPy for image processing and Matplotlib for displaying the results.

9.1 Implementation Steps

Input Image: A remote sensing image is loaded using OpenCV.

Preprocessing: The image is resized if required and converted to grayscale for easier pixel analysis.

Region Analysis: The mean and standard deviation of each region are calculated.

Quadtrees Splitting: If the standard deviation is greater than the selected threshold, the region is divided into four smaller regions. This process continues recursively.

Region Merging: Neighboring regions with similar mean values are identified and merged using the merging threshold.

Terrain Classification: The final regions are classified into categories such as vegetation, water, urban areas, and bare land.

Output: The original and segmented images are displayed for comparison.

9.2 Implementation Flow

Input Image → Preprocessing → Region Analysis → Quadtree Splitting → Region Merging → Terrain Classification → Final Output

10.RESULTS AND DISCUSSION

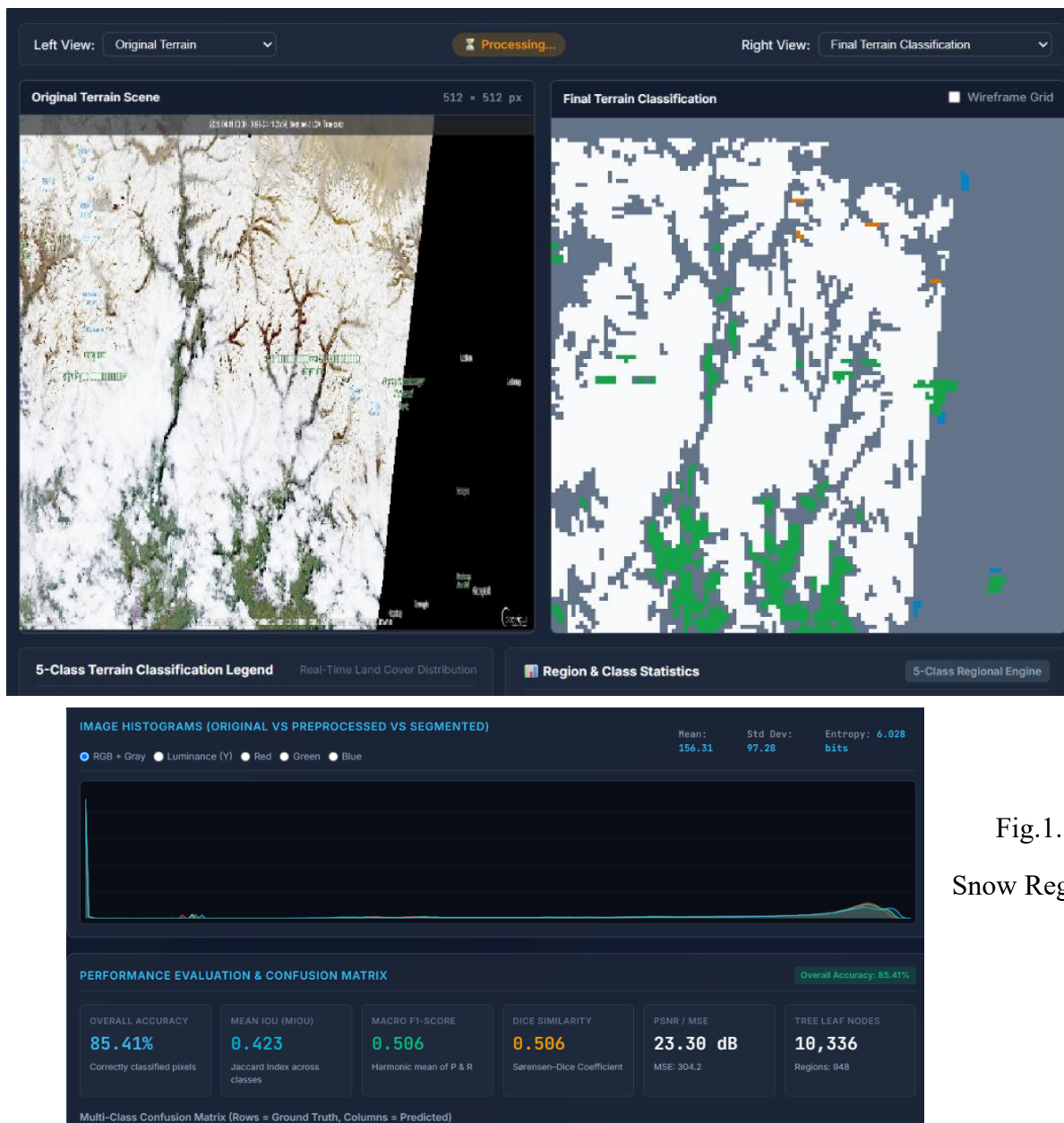


Fig.1.
Snow Region

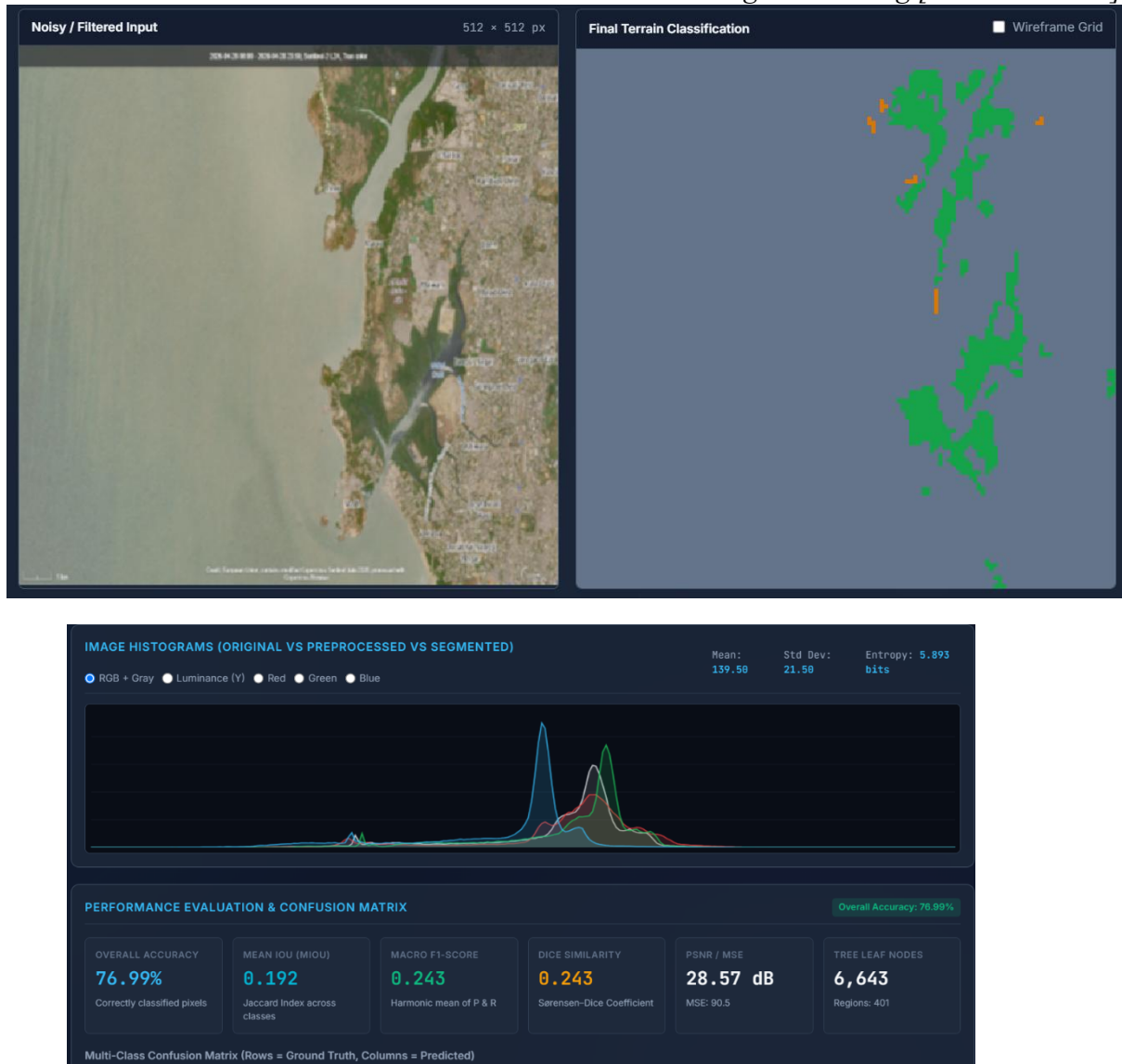
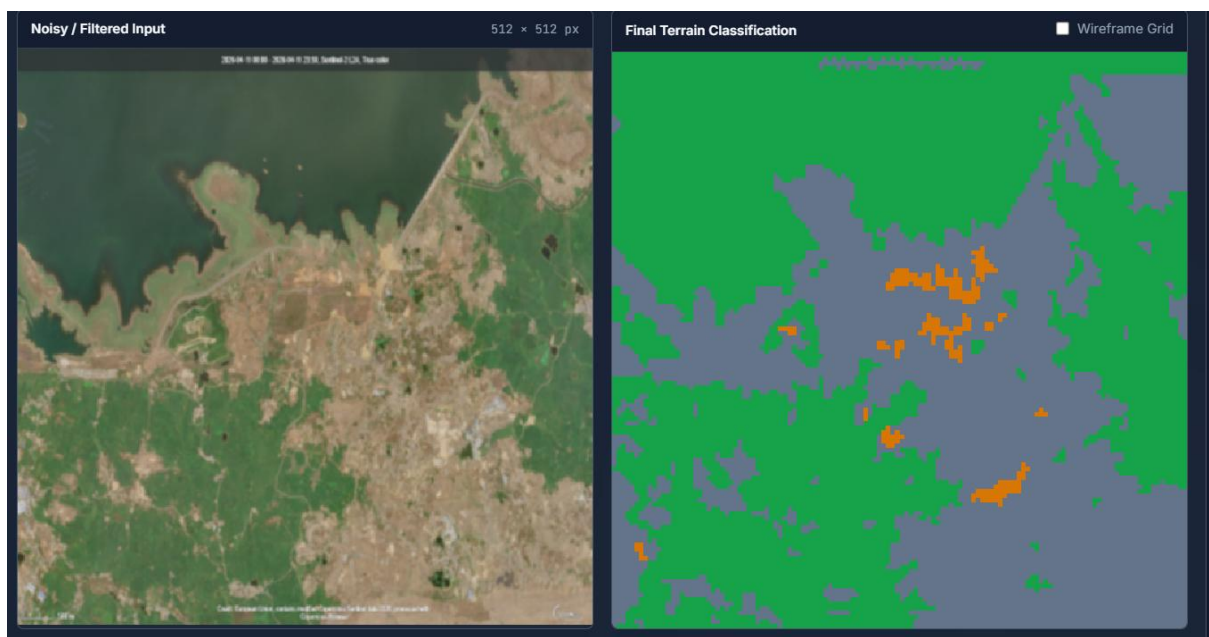


Fig. 2. Barran Land



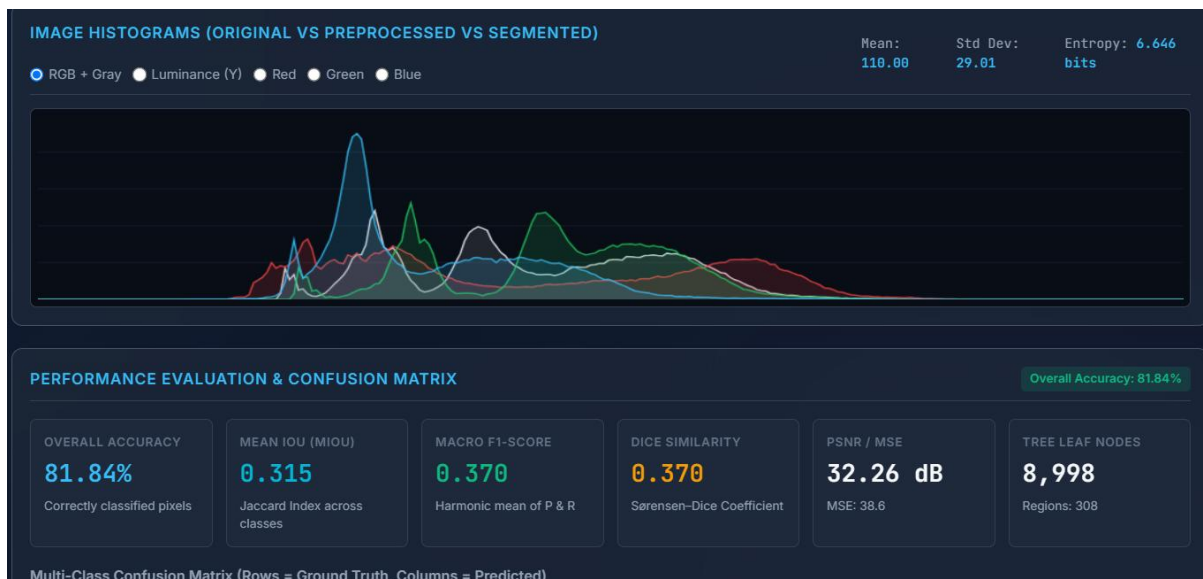


Fig. 3. Dense Forest

The project includes five 512×512 procedural terrain scenes. The reported baseline uses $T_s=14$, $T_m=14$, $T_{edge}=35$ and $A_{min}=36$ pixels. The values below are the benchmark results recorded in the supplied project report.

11.COCLUSION AND FUTURE SCOPE

The project successfully demonstrates terrain classification using **Quadtree Region Splitting and Region Adjacency Graph (RAG) based Region Merging**. The system performs preprocessing, filtering, segmentation and classification, while reducing unnecessary fragmented regions. The web interface also allows users to observe the effect of different noise types, filters and kernel sizes. The reported results show good performance, with a **macro-average accuracy of 95.51%**.

Future Scope

- Testing the system on **real satellite and remote-sensing images**.
- Adding **machine learning/deep learning** for improved terrain classification.
- Including **texture and multispectral features** for better class distinction.
- Adding more **terrain and land-cover classes**.
- Optimizing the system for **faster processing of large images**.
- Developing the system for practical applications such as **agriculture monitoring, land-use mapping and water-body detection**.

12. REFERENCES

- R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed. New York, NY, USA: Pearson, 2018.
- S. L. Horowitz and T. Pavlidis, "Picture segmentation by a tree traversal algorithm," Journal of the ACM (JACM), vol. 23, no. 2, pp. 368–388, Apr. 1976.
- R. M. Haralick and L. G. Shapiro, "Image segmentation techniques," Computer Vision, Graphics, and Image Processing, vol. 29, no. 1, pp. 100–132, Jan. 1985.
- C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in Proc. 6th IEEE Int. Conf. Computer Vision (ICCV), Bombay, India, 1998, pp. 839–846.
- N. Otsu, "A threshold selection method from gray-level histograms," IEEE Trans. Systems, Man, and Cybernetics, vol. SMC-9, no. 1, pp. 62–66, Jan. 1979.
- G. M. Foody, "Status of land cover classification accuracy assessment," Remote Sensing of Environment, vol. 80, no. 1, pp. 185–201, Apr. 2002.
- World Wide Web Consortium (W3C), "HTML5 2D Canvas Context Specification," W3C Recommendation, 2021. [Online]. Available: <https://www.w3.org/TR/2dcontext/>

13. APPENDIX

Key JavaScript algorithmic implementations from app.js are documented below:

```
// 1. Quadtree Recursive Splitting Predicate (app.js)
function canSplitBlock(node, Ts, minBlock) {
  if (node.size <= minBlock) return false;
  return (node.stdDevLum > Ts || node.rangeLum > Ts * 2.6 || node.meanEdge > Ts * 1.8);
}

// 2. Class-Aware RAG Merge with Edge Gradient Inhibition
function shouldMergeRegions(u, v, Tm, colorWeight, edgeMap, Tedge) {
  const contactEdgeGrad = computeBoundaryGradient(u, v, edgeMap);
  if (contactEdgeGrad > Tedge) return false; // Inhibit merge across sharp boundaries
  const spectralDist = (1 - colorWeight) * Math.abs(u.meanLum - v.meanLum) +
    colorWeight * (euclideanDist(u.meanRGB, v.meanRGB) / Math.sqrt(3));
  if (u.predictedClass === v.predictedClass) return spectralDist <= (Tm * 2.2);
  return spectralDist <= Tm;
}

// 3. Consolidated 5-Class Terrain Spectral Classifier
function predictRegionClass(reg) {
  const { meanR, meanG, meanB, hsv, gi, wi } = reg.features;
  if (hsv.v > 0.84 && hsv.s < 0.18) return 2; // Snow (Pure White)
  if (meanB > meanG * 1.15 && meanB > meanR * 1.25) return 0; // River / Water
  if (hsv.h >= 22 && hsv.h <= 55 && hsv.s >= 0.25) return 1; // Desert / Sand
  if (gi > 0.035 || (hsv.h >= 65 && hsv.h <= 165 && hsv.s > 0.16)) return 3; // Forest / Agriculture
```

```

return 4; // Urban / Barren Land
}

```

5 terrain classes:

Class	Purpose in project	Displayed class color
River / Water	Water bodies, rivers and coastal water	Blue
Desert / Sand	Sand, dunes and yellow/brown dry terrain	Amber
Snow	Bright, low-saturation snow/glacial regions	White
Forest / Agriculture	Vegetation, forest and agricultural green regions	Green
Urban / Barren	Built-up, rock, exposed soil and neutral terrain	Slate gray

Scene	Accuracy	Macro F1	Dice	PSNR dB	Leaves	Merged	Reduction	Time
Desert Dunes	98.94%	0.848	0.848	38.7	1,498	32	97.9%	56 ms
Dense Forest	99.39%	0.980	0.980	35.7	2,206	38	98.3%	69 ms
Alpine Peaks & Snow	88.71%	0.806	0.806	30.5	11,047	514	95.3%	288 ms
Farmland Mosaic	91.75%	0.854	0.854	24.5	8,248	804	90.3%	150 ms
Coastal Estuary	98.78%	0.888	0.888	35.3	2,362	27	98.9%	49 ms
Macro average	95.51%	0.875	0.875	32.9	25,361	1,415	94.4%	122 ms

Fig. Performance Table